



Name: M. Nivedha

Reg No: 192471040

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Course: Machine Learning

Assessment: Col-AT1-Application

Based Question

A factory has three boxes containing colored balls

Box A contains 6 red, 4 blue and 2 green balls;

and Box B contains 5 red, 7 blue, and 3 green balls;

and Box C contains 8 red, 2 blue and 5 green balls.

Box B is twice as likely to be selected as Box A or

Box C. if a ^{Random} ~~Normaly~~ drawn ball is Blue, determine the

probability that it came from Box C using Bayes

theorem

Box	Red	Blue	Green	Total
A	6	4	2	12
B	5	7	3	15
C	8	2	5	15

Since Box B is twice as likely to be selected:

$$P(A) = \frac{1}{4}, P(B) = \frac{1}{2}, P(C) = \frac{1}{4}$$

Probability of drawing a blue ball.

$$P(\text{Blue} | A) = \frac{4}{12} = \frac{1}{3}$$

$$P(\text{Blue} | B) = \frac{7}{15}$$

$$P(\text{Blue} | C) = \frac{2}{15}$$

Using Bayes theorem,

$$P(C | \text{Blue}) = \frac{P(\text{Blue} | C) P(C)}{P(\text{Blue})}$$

$$= \frac{2}{15} \times \frac{1}{4}$$

$$\frac{1}{3} \times \frac{1}{4} + \frac{7}{15} \times \frac{1}{2} + \frac{2}{15} \times \frac{1}{4}$$

$$= \frac{\frac{1}{30}}{\frac{17}{30}} = \frac{2}{17}$$

$$P(C | \text{Blue}) = \frac{2}{17} = 0.1176 = 11.76\%$$

Therefore, the probability that the selected box was Box C is 11.76%.

parking faculty has three sections containing cars
 of different colors. Section A has 3 red, 3 blue and 2
 black cars; section B has 4 red, 6 blue and 5 black
 cars; section C has 7 red, 2 blue and 6 black cars.
 Section B is twice as likely to be selected as
 section A and C. If a randomly selected car is blue, determine the
 probability that it came from section C using Bayes's
 theorem

Section	Red	Blue	Black	Total
A	3	3	2	8
B	4	6	5	15
C	7	2	6	15

Since section B is twice as likely to be selected.

$$p(A) = \frac{1}{4}, p(B) = \frac{1}{2}, p(C) = \frac{1}{4}$$

probabilities of selecting a blue car:

$$p(\text{Blue} | A) = \frac{3}{10}$$

$$p(\text{Blue} | B) = \frac{1}{15} = \frac{2}{5}$$

$$p(\text{Blue} | C) = \frac{2}{15}$$

Using Bayes theorem,

$$p(C | \text{Blue}) = \frac{p(\text{Blue} | C) p(C)}{p(\text{Blue})}$$

$$= \frac{\frac{2}{15} \times \frac{1}{4}}{\frac{3}{10} \times \frac{1}{4} + \frac{2}{5} \times \frac{1}{2} + \frac{2}{15} \times \frac{1}{4}}$$

$$= \frac{1}{\frac{30}{11} \times \frac{1}{60}} = \frac{2}{11}$$

$$= \frac{1}{\frac{30}{11} \times \frac{1}{60}} = \frac{2}{11}$$

$$p(C | \text{Blue}) = \frac{2}{11} = 0.1818 = 18.18\%$$

probability that a blue car came from section C is 18.18%

A machine learning model is used to classify breast cancer cases as malignant or benign. The dataset contains 1800 samples from each class, and 20% of the total data is used for testing. The model produces 210 true positives, 190 true negatives and 30 false positives. Determine the False Negative (FN) and calculate the Accuracy, precision, sensitivity (Recall) and specificity of the model. Briefly interpret the results for medical diagnosis.

Total samples:

$$1800 + 1800 = 3600$$

Testing data:- $3600 \times 20\% = 720$

Given: $TP = 210$

$$TN = 190$$

$$FP = 30$$

False Negative:

$$FN = 720 - (210 + 190 + 30)$$

$$FN = 290$$

Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \times 100$$

$$= \frac{210 + 190}{720} \times 100$$

$$\text{Accuracy} = 55.56\%$$

Precision

$$\text{Precision} = \frac{TP}{TP + FP} \times 100$$

$$= \frac{210}{210 + 30} \times 100$$

$$\text{Precision} = 87.5\%$$

Sensitivity / Recall

$$\text{Sensitivity} = \frac{TP}{TP + FN} \times 100$$

$$= \frac{210}{210 + 290} \times 100$$

$$\text{Sensitivity} = 42\%$$

Specificity

$$\text{Specificity} = \frac{TN}{TN + FP} \times 100$$

$$= \frac{190}{190 + 30} \times 100$$

$$\text{Specificity} = 86.36\%$$

Interpretation:-

The model has high precision and specificity, but its sensitivity is only 42%. Therefore, it misses a considerable number of malignant cases. For medical diagnosis, improving sensitivity is important because false negative can cause actual cancer cases to be missed.

machine learning model shows the following
training and validation losses over eight epochs
training loss = 2.8, 2.5, 2.2, 2.0, 1.8, 1.6, 1.5, 1.3 and
Validation loss 2.4, 2.2, 2.0, 2.1, 2.3, 2.6, 2.9, 3.2

Identify the epoch at which overfitting begins.

Suggest two techniques to reduce overfitting
and explain its effect on the model's ability to
generalise to unseen data.

1) Epoch at which overfitting begins?

The validation loss decreases until Epoch 3, where it
reaches its lowest value of 2.0. From Epoch 4, validation
loss starts increasing while training loss continues
decreasing.

Overfitting begins at Epoch 4

ii) Two techniques to reduce overfitting.

i) Early stopping:-

Training can be stopped when validation loss starts increasing. This prevents the model from learning unnecessary patterns from the training data.

2) Regularization:-

L1 or L2 Regularization can be added to reduce model complexity and prevent the model from depending too much on particular features.

iii) Effect on generalization

Overfitting occurs when the model learns the training data too closely, including noise and unnecessary patterns. Although the training loss continues to decrease, the training to validation loss increases.

Therefore, the model performs well on training data but performs poorly on unseen data.

5) A Machine-learning model has to follow ^{minimum} loss. The model should preferably be stopped around Epoch 3, where the Validation loss is minimum.

Training and Validation accuracies over eight epochs:

Training Accuracy = 60%, 65%, 75%, 70%, 80%, 85%, 88%, 92%.

Validation Accuracy = 58%, 63%, 68%, 72%, 73%, 72%, 70%, 68%.

Identify the epoch at which overfitting starts.

Explain why validation accuracy decreases while

training accuracy increases and suggest two methods

to improve validation performance.

i) Epoch at which overfitting starts

Validation accuracy increases up to Epoch 5, reaching

73%. After Epoch 5, it decreases while training

data. After some point, the model starts memorizing

specific training examples rather than learning

general patterns. As a result, its performance on

unseen validation data decreases.

Thus:

- * Training accuracy $\rightarrow$ Increase

- * Validation Accuracy $\rightarrow$ decrease

- * This indicates overfitting

i) Two methods to improve validation performance

i) Dropout:

Some neurons are randomly disabled during

training. This reduces dependence on particular

neurons and improves generalization.

stopping:-

training can be stopped when validation accuracy stops improving. Here the highest validation accuracy is 73% at Epoch 5, so training can be stopped around this point.

Conclusion:-

The decrease in validation accuracy after Epoch 5 indicates overfitting, while training accuracy continues to rise.